



CSE330 Numerical Methods Syllabus Spring 2023 - Abu Mohammad Khan

Computer Programing (Java-1) (BRAC University)



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A. Course General Information:

Course Code:	CSE330
Course Title:	Numerical Methods
Credit Hours (Theory+Lab):	3+0
Contact Hours (Theory+Lab):	3+3
Category:	Program Core
Type:	Required
Prerequisites:	MAT216
Co-requisites:	None

B. Course Catalog Description (Content):

Fixed Point Arithmetic, Polynomial Interpolation, Differentiation, Nonlinear Equations, Linear Equations, Least-Squares Approximation, Numerical Integration

C. Course Objective:

The course will help students to recognize the need for numerical analysis, and importance of error analysis. They will learn various methods to linearize a polynomial, differentiate and integrate different functions by using approximations, and finally how to solve the linearized equation by using the laws of linear algebra, like Gaussian elimination, QR decomposition, etc.

D. Faculty Information:

Faculty Name	Faculty Initials	Section(s) teaching
Abu Mohammad Khan	AMK	1
Rafeed Rahman	RRH	5
Aquib Azmain	MAA	3, 12
Avijit Biswas	ABW	8, 9, 10
Shoumma Ahsan Khondokar	SAK	11
Jawaril Munsad Abedin	JMA	2
Zahin Wahab	ZWA	4, 7

E. Course Outcomes (COs):

Upon successful completion of this course, students will be able to

Sl.	CO Description	Weightage (%)
CO1	Demonstrate an understanding of the fundamental concept of numerical analysis for different mathematical problems and the need for error analysis.	10
CO2	Comprehend different numerical techniques and theorems of numerical methods for polynomial interpolation, derivatives, linear and non-linear equations, and integrations.	20
CO3	Apply the principles and/or techniques of numerical methods in different mathematical setups.	40
CO4	Solve the linear and overdetermined systems by different numerical methods. Also compute integration numerically using Newton-Cotes formula and estimate the error.	30

F. Mapping of CO-PO-Taxonomy Domain & Level- Delivery-Assessment Tool:

Sl.	CO Description	POs	Bloom's taxonomy domain/level	Delivery methods and activities	Assessment tools
CO1	Demonstrate an understanding of the fundamental concept of numerical analysis for different mathematical problems and the need for error analysis.	(b)	Cognitive/Analyze	Lectures, Notes, Lab	Assignments, Quiz, Midterm, Final exam
CO2	Comprehend different numerical techniques and theorems of numerical methods for polynomial interpolation, derivatives, linear and non-linear equations, and integrations.	(c)	Cognitive/Evaluate	Lectures, Notes, Lab	Assignments, Quiz, Midterm, Final exam
CO3	Apply the principles and/or techniques of numerical methods in different mathematical setups.	(a)	Cognitive/Analyze	Lectures, Notes, Lab	Assignments, Quiz, Midterm, Final exam
CO4	Solve the linear and overdetermined systems by different numerical methods. Also compute integration numerically using Newton-Cotes	(a)	Cognitive/Analyze	Lectures, Notes,	Assignments, Quiz, Midterm,

	formula and estimate the error.			Lab	Final exam
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G. Course Materials:

i. Text and Reference Books:

Sl .	Title	Author(s)	Publication Year	Edition	Publisher	ISBN
1.	Numerical Analysis II – Lecture Notes. (Main Text)	Anthony Yeates.	2018		Durham University	
2.	Numerical Methods In Engineering With Python.	Jaan Kiusalaas.	2005	First Edition	Cambridge University Press	ISBN-13: 978-0-521-85287-6 ISBN-10:0-521-85287-0
3.	Numerical Analysis (Reference)	Richard L. Burden J. Douglas Faires	2011	Ninth Edition	BROOKS/ COLE CENGAGE Learning	ISBN-13:978-0-538-73351-9 ISBN-10:0-538-73351-9

ii. Other materials (if any)

(a) Lecture Notes/Handouts

H. Lesson Plan:

Day	:	Topics
Week # 1	:	Introduction. Overview of the whole course
	:	Floating point Arithmetic
Week# 2	:	Rounding error loss of significance. Quiz -1. Assignment-1.
	:	Polynomial Interpolation
Week # 3	:	Lagrange Form
	:	Newton divided/difference form.
Week # 4	:	Interpolation error, convergence. Quiz -2. Assignment-2.
	:	Derivative condition, Hermite basis
Week # 5	:	Differentiation. Higher order finite difference
	:	Rounding error. Richardson Extrapolation.
Week # 6	:	Nonlinear equations: Interval bisection method. Quiz -3. Assignment-3.
	:	Fixed point iteration, order of convergence
Week # 7	:	MIDTERM WEEK. NO CLASS
	:	March 05, 2023 (Sunday) from 2:00pm-4:00pm
Week # 8	:	Newton's method, Aitken acceleration.
	:	Quasi-Newton method (Secant form)

Week # 9	:	Linear equations. Triangular forms. Quiz -4. Assignment-4.
	:	Gaussian eliminations method
Week # 10	:	LU decomposition method
	:	Pivoting, Vector norms, Matrix norms
Week # 11	:	Least Square Approximation: Orthogonality.
	:	Discrete least squares. Quiz -5. Assignment-5.
Week # 12	:	QR-decomposition
	:	Newton-Cotes Formulae.
Week # 13	:	Composite Newton-Cotes formulae
	:	Exactness
Week # 14	:	Gaussian Quadrature
	:	Reviews. Quiz -6. Assignment-6.
Final Exam	:	May 03, 2023 (Wednesday) from 2:00pm-4:00pm

I. Assessment Tools:

Assessment Tools	Weightage (%)
Attendance	5
Quiz	10
Assignment	15
Midterm	20
The mandatory final exam	30
Lab	20

J. CO Assessment Plan:

Assessment Tools	Course Outcomes			
	CO1	CO2	CO3	CO4
Assignments	√	√	√	√
Quiz	√	√	√	√
Mid Term	√	√	√	√
Final	√	√	√	√

K. CO Attainment Policy:

As per BRAC University Policy.

L. Grading policy:

As per BRAC University Policy.

M. Course Coordinator:
Abu Mohammad Khan (AMK)